

Combinational Logic Circuits

Lecture 8

Design Methods

Type of IC chips (based on packing density) :

- Small-scale integration (SSI): up to 12 gates
- Medium-scale integration (MSI): 12-99 gates
- Large-scale integration (LSI): 100-9999 gates
- Very large-scale integration (VLSI): 10,000-99,999 gates
- Ultra large-scale integration (ULSI): > 100,000 gates



Design Methods

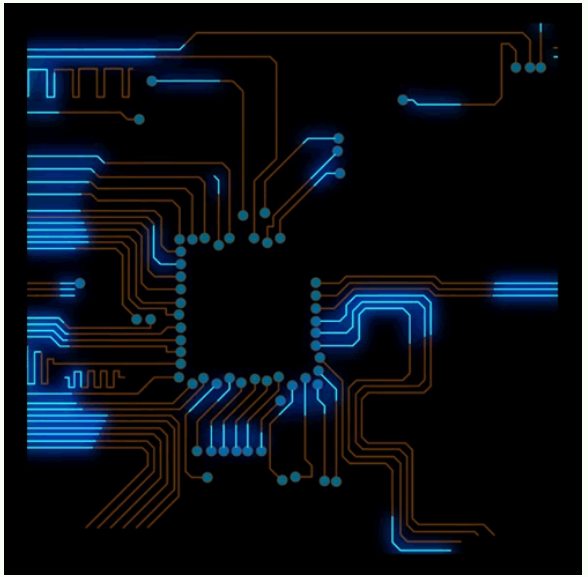
Main objectives of circuit design:

1. Reduce cost

   - reduce number of gates
   - reduce IC packages

2. Increase speed

3. Design simplicity (reuse blocks where possible)



Logic circuits

1. Combinational

Outputs depend only on the current inputs.

Examples:

- Encoder
- Decoder
- Adders
- Subtracters
- Multiplexer
- Demultiplexer

2. Sequential

Outputs depend on the current inputs and previous inputs.

Examples:

- Flip-flops
- Counters
- Registers
- Clocks

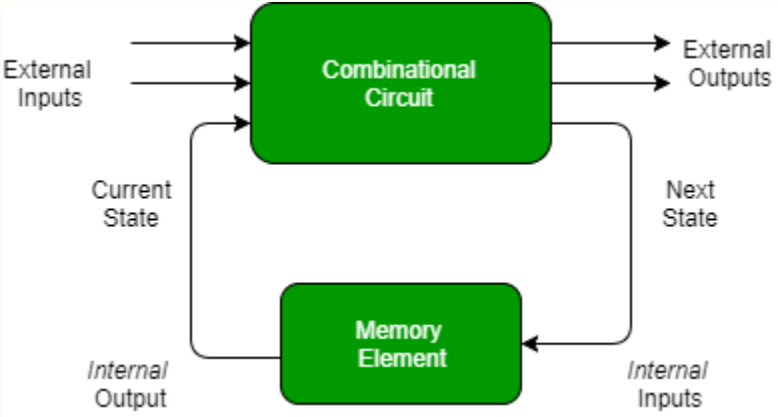
Logic circuits

1. Combinational



Figure: Combinational Circuits

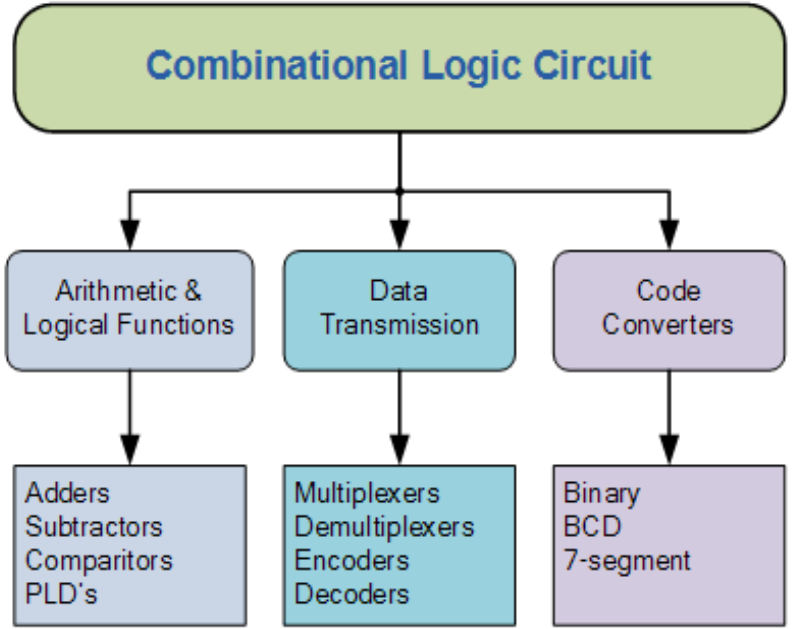
2. Sequential



Combinational	Sequential
• In this output depends only upon present input.	• In this output depends upon present as well as past input.
• Speed is fast.	• Speed is slow.
• It is designed easy.	• It is designed tough as compared to combinational circuits.
• There is no feedback between input and output.	• There exists a feedback path between input and output.
• This is time independent.	• This is time dependent.
• Elementary building blocks: Logic gates	• Elementary building blocks: Flip-flops
• Used for arithmetic as well as boolean operations.	• Mainly used for storing data.
• Don’t have capability to store any state.	• Have capability to store any state or to retain earlier state.
• These circuits do not have any memory element.	• These circuits have memory element.
• It is easy to use and handle.	• It is not easy to use and handle.

Applications of Combinational Logic Circuits

- Calculators
- Digital measuring techniques
- Computers
- Digital processing
- Automatic control of machines
- Industrial processing
- Digital communications



Design Combinational Logic Circuits

Steps

1. Identify Problem from Verbal Description.
2. Specify the number of inputs and required numbers of outputs and assign symbols.
3. Construct the truth table to define relationship between inputs and outputs.
4. Simplify Boolean function using K-Map or Boolean Algebra rules.
5. Draw logic diagram for the simplified logic expression.

Q: Design combinational logic circuit for the below scenario

A committee of three individuals decide proposal for an organization. Each individual votes either yes or no for each proposal that arises. A proposal is passed if it receives at least two yes votes. Design a circuit that determines whether a proposal passes.



Solution

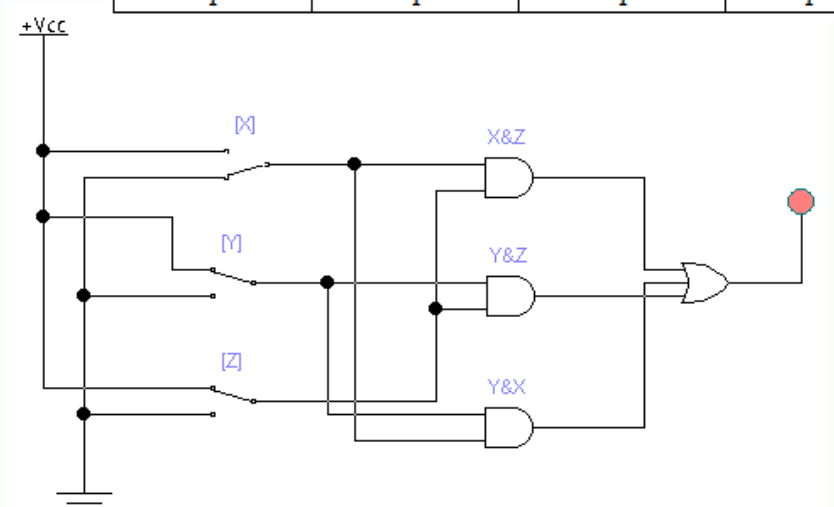
Three inputs (x, y, z) – Individual votes

One Output (F) – Whether proposal passed or not

Individual1 (X)	Individual2 (Y)	Individual3 (Z)	Proposal(F)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

X \ YZ	00	01	11	10
0			1	
1		1	1	1

$$F = XY + XZ + YZ$$



Encoder

- A device that codes (converts) the data in one format to another.
- Encoding is done for variety of purposes in practical applications.

  E.g., Security, Efficiency, Speed, Minimal Design

- 2 Main types are,

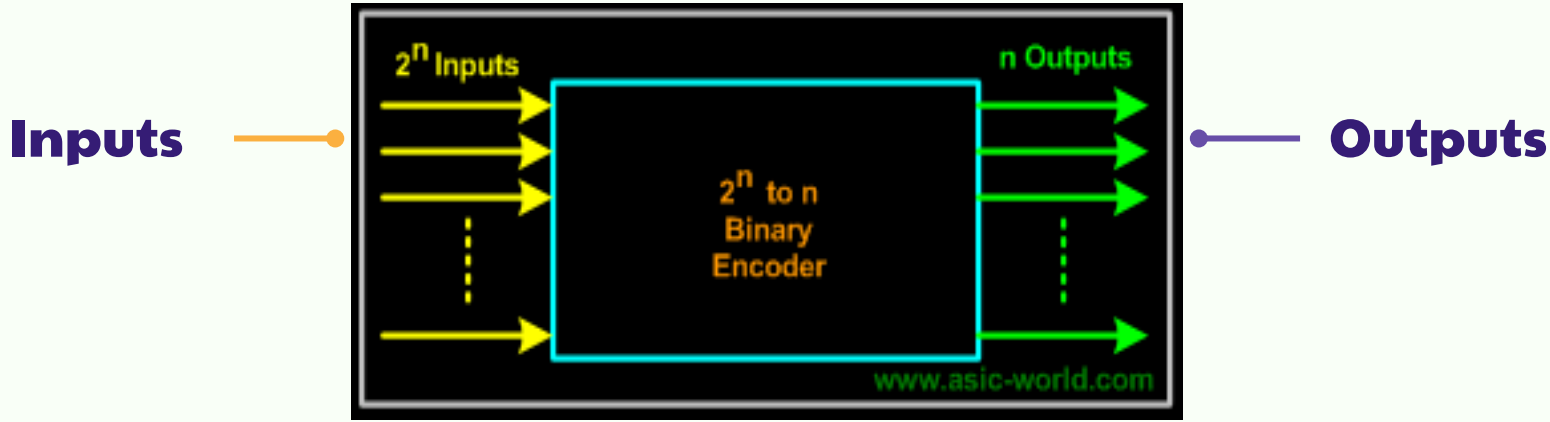
  1. Binary Encoder

  2. Priority Encoder

Binary Encoder

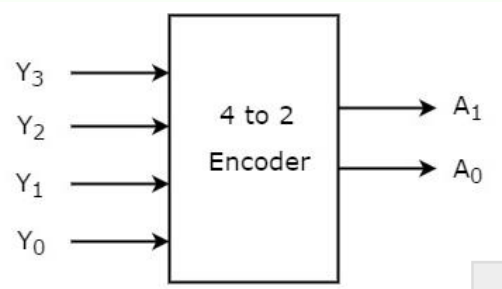
An encoder is a combinational circuit that converts binary information in the form of a

2^n input lines into N output lines



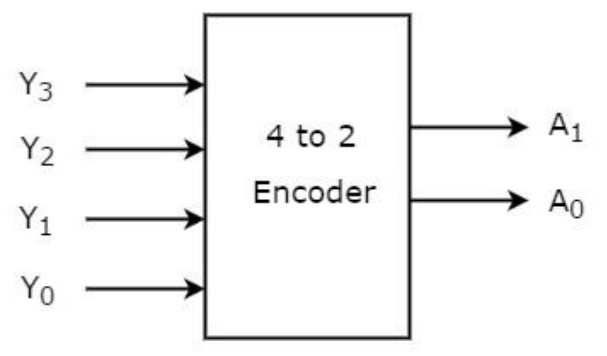
Limitation - Only one input can be active at any given time. It would produce an error if more than one input are active at the same time.

Binary Encoder : 4 x 2 Encoder



Inputs				Outputs	
Y_3	Y_2	Y_1	Y_0	A_1	A_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

Binary Encoder : 4 x 2 Encoder

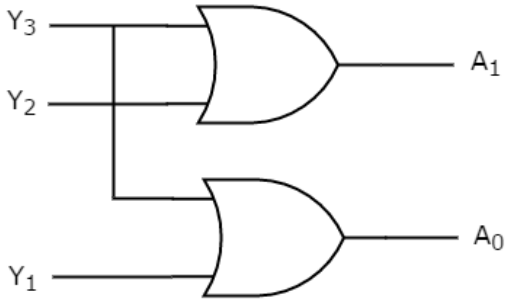


Inputs				Outputs	
Y_3	Y_2	Y_1	Y_0	A_1	A_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

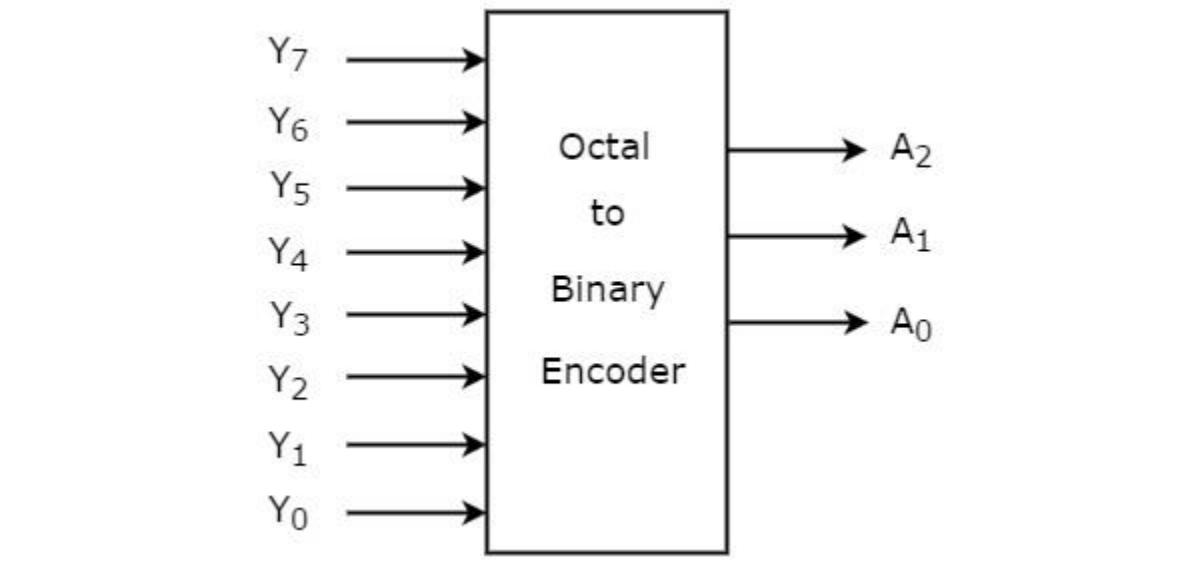
Boolean Expression for,

$$A_1 = Y_3 + Y_2$$

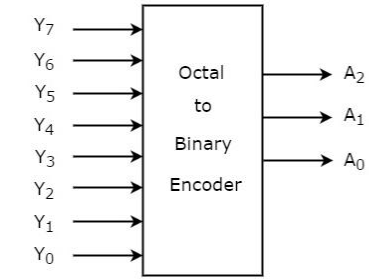
$$A_0 = Y_3 + Y_1$$



Q: Give a try : 8 x 3 Encoder



Binary Encoder : 8 x 3 Encoder



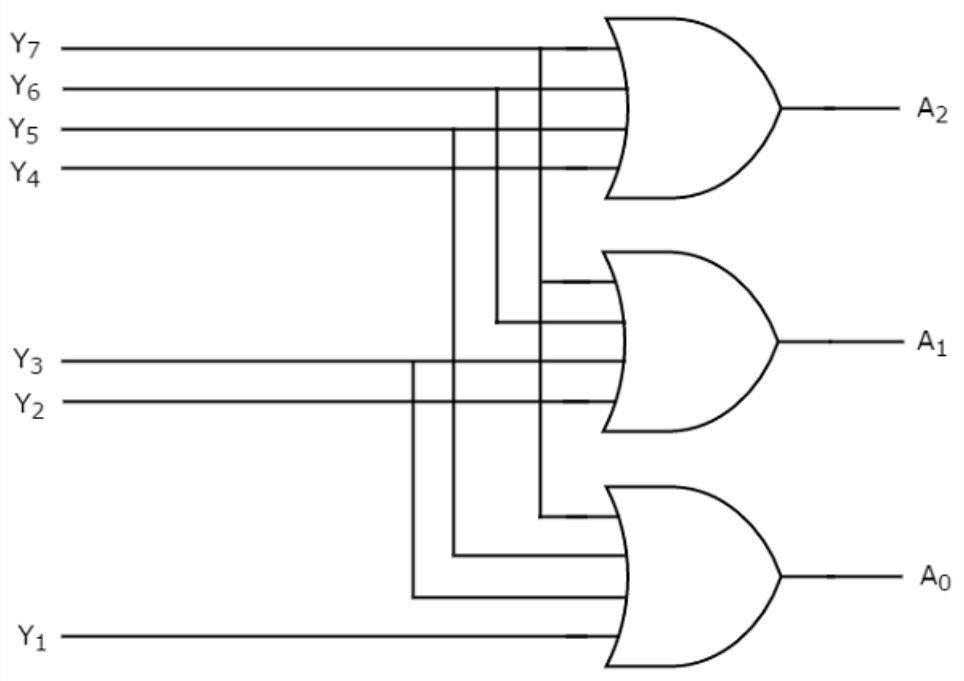
Inputs								Outputs		
Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Binary Encoder : 8 x 3 Encoder

$$A_2 = Y_7 + Y_6 + Y_5 + Y_4$$

$$A_1 = Y_7 + Y_6 + Y_3 + Y_2$$

$$A_0 = Y_7 + Y_5 + Y_3 + Y_1$$

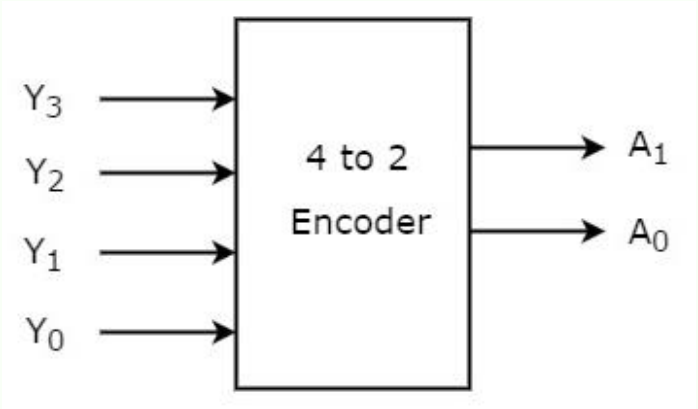


Priority Encoders

- If more than one input is active high, then the binary encoder produces an output, which may not be the correct code.
- The Priority Encoder solves the problems mentioned above by allocating a priority level to each input.
- In priority encoders, more than one input can be high at any given time.
- Output corresponds to the currently active input which has the highest priority. So, when an input with a higher priority is present, all other inputs with a lower priority will be ignored

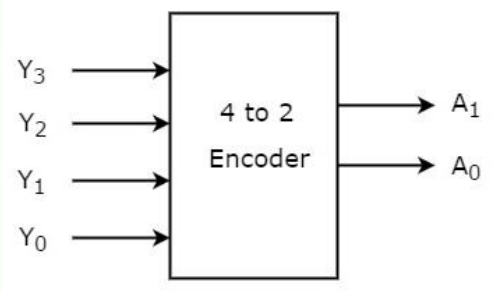
4 x 2 Priority Encoders

Here, the input, Y3 has the highest priority, whereas the input, Y0 has the lowest priority. In this case, even if more than one input is ‘1’ at the same time, the output will be the binary code corresponding to the input, which is having higher priority.



4 x 2 Priority Encoders

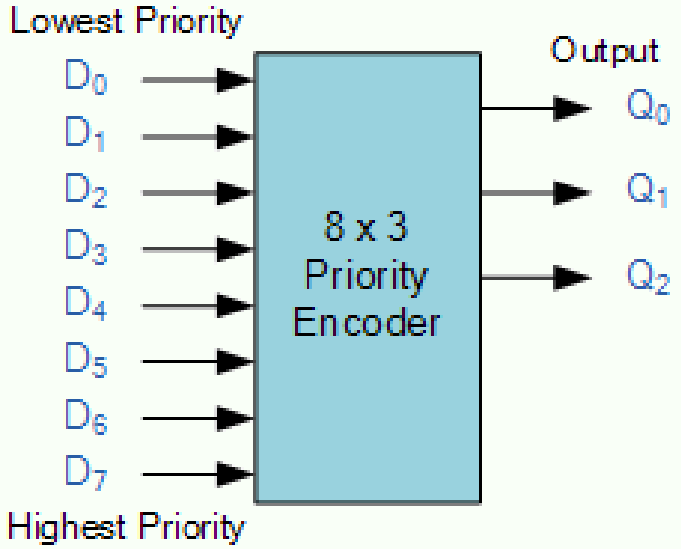
Here, the input, Y3 has the highest priority, whereas the input, Y0 has the lowest priority. In this case, even if more than one input is ‘1’ at the same time, the output will be the binary code corresponding to the input, which is having higher priority.



Inputs				Outputs	
Y_3	Y_2	Y_1	Y_0	A_1	A_0
0	0	0	1	0	0
0	0	1	x	0	1
0	1	x	x	1	0
1	x	x	x	1	1

8 X 3 Priority Encoders

D7 has the highest priority, whereas the input, D0 has the lowest priority.

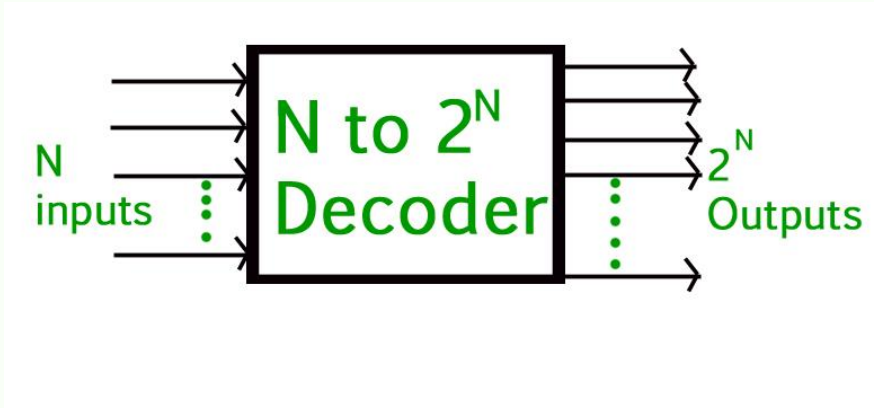


Inputs								Outputs		
D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Q_2	Q_1	Q_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	x	0	0	1
0	0	0	0	0	1	x	x	0	1	0
0	0	0	0	1	x	x	x	0	1	1
0	0	0	1	x	x	x	x	1	0	0
0	0	1	x	x	x	x	x	1	0	1
0	1	x	x	x	x	x	x	1	1	0
1	x	x	x	x	x	x	x	1	1	1

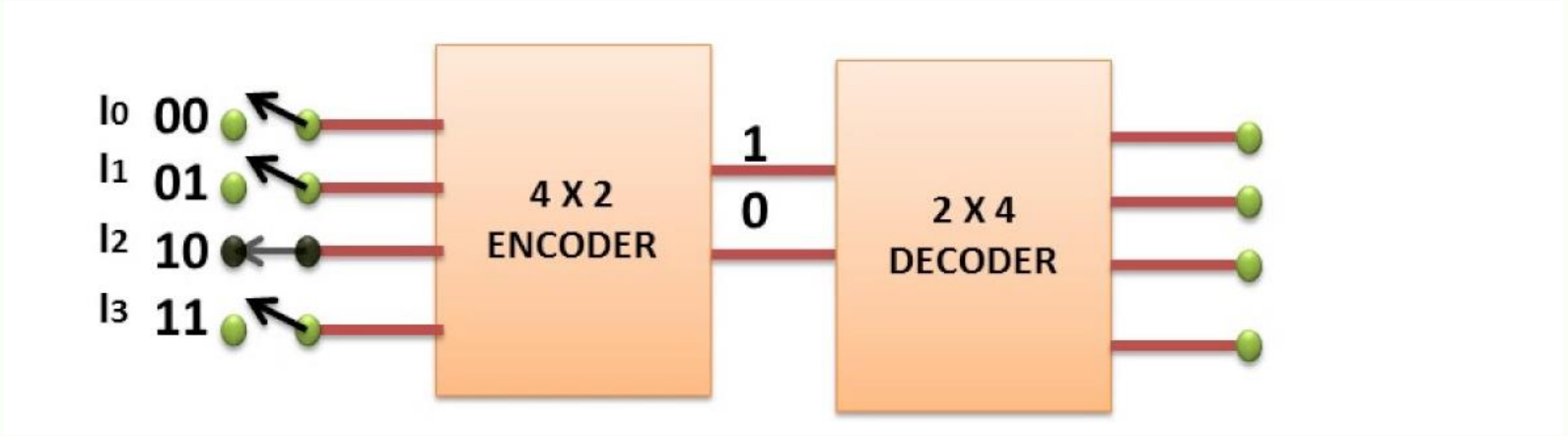
X = dont care

Decoder

- Once the information is encoded (at the transmitting side) it must be decoded later (at the receiving side).
- Translate or decode coded information from one format into another
- An encoder is a combinational circuit that converts ‘n’ binary input signals into an equivalent code using 2^n outputs.

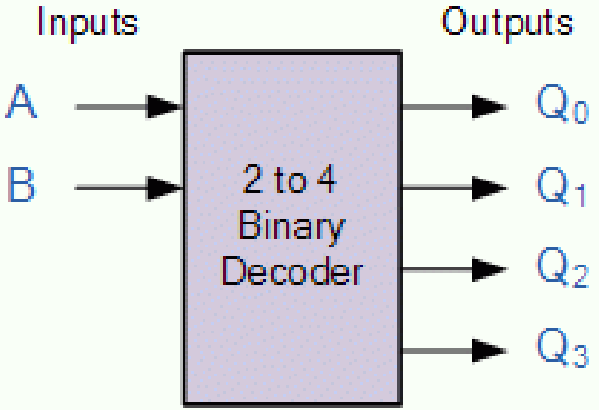


Decoder



Sender **Receiver**

2 x 4 Decoder



Truth Table

A	B	Q_0	Q_1	Q_2	Q_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

The output values will be:

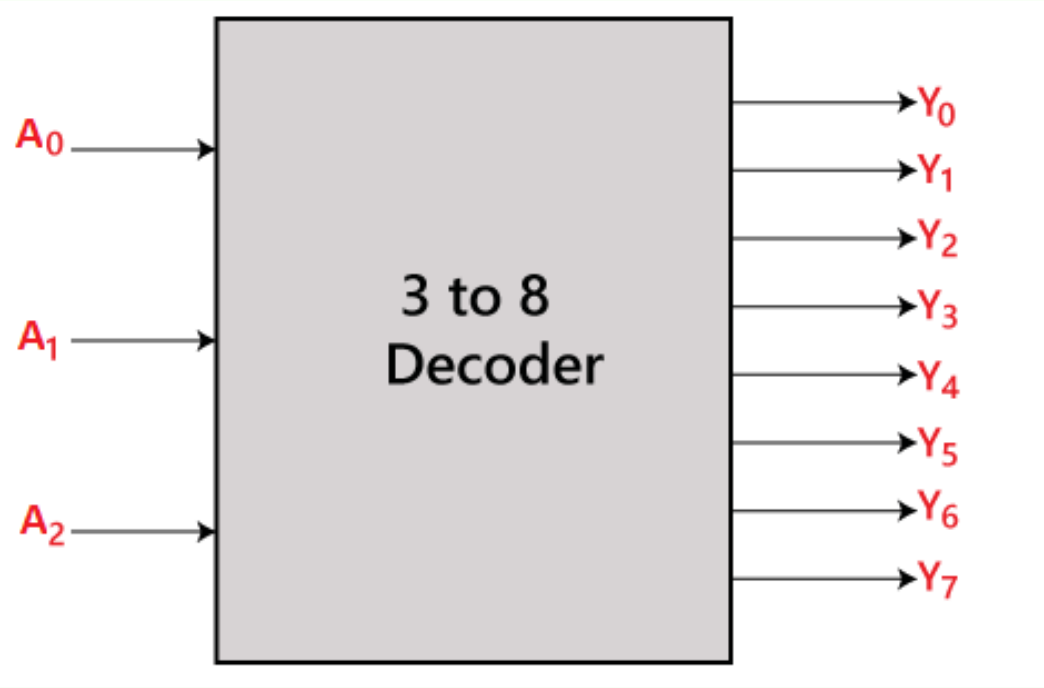
Qo=A'B'

Q1=A'B

Q2=AB'

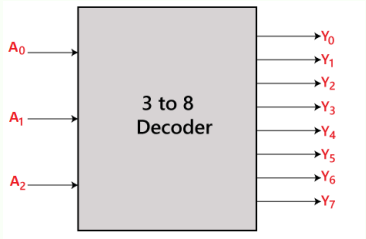
Q3=AB

8 x 3 Decoder



8 x 3 Decoder

INPUTS			Outputs							
A_2	A_1	A_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0



$Y_0 = A_0'.A_1'.A_2'$

$Y_1 = A_0.A_1'.A_2'$

$Y_2 = A_0'.A_1.A_2'$

$Y_3 = A_0.A_1.A_2'$

$Y_4 = A_0'.A_1'.A_2$

$Y_5 = A_0.A_1'.A_2$

$Y_6 = A_0'.A_1.A_2$

$Y_7 = A_0.A_1.A_2$

Thanks!

Do you have any questions?

Type your text

